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Virtual Assistant in Pediatric Hematology & Oncology

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> Current Situation

Clinicians and researchers in pediatric hematology and oncology face a time-consuming process for retrieving critical medical information:

- **Manual and Time-Consuming:** Practitioners rely on a manual search process to sift through thousands of pages of clinical trial protocols and institutional documents to find specific medical criteria or treatment details. This manual search is made even more difficult because the protocols may contain charts, images, and cross-references to each others.
- **Fragmented Information:** While the BSPHO website hosts the most frequently used treatment guidelines, individual hospitals and centers often have their own specific guidelines that are not centralized on the site.
- **Constant Updates:** These medical documents frequently receive amendments and need to be continuously updated, making it difficult to ensure practitioners are looking at the most current information.

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> Tool to build: Must-haves Features & Requirements (1/2)

An intelligent search tool to navigate thousands of pages of protocols and institutional documents, allowing clinicians and researchers to find critical information quickly and accurately.

- **Grounding & Citations:** the UX features inline citations with the document title and page number.
- **Conversational Search / Workflow:** the tool may proactively ask clarifying questions before answering. Eg: if the response to a user's question varies based on patient risk groups, on patient ages, or on other factors.
- **Strict Guardrails:** The tool is restricted to acting purely as a search utility, not a medical device. It must strictly use a professional, medical tone, base all medical facts on the provided English BSPHO protocols, and explicitly state if an answer is not covered rather than guessing or making up facts.
- **(Admin) Knowledge Management:** A user-friendly admin workflow allows for the easy upload of new PDF protocols and amendments, ensuring the knowledge base remains continuously up-to-date.

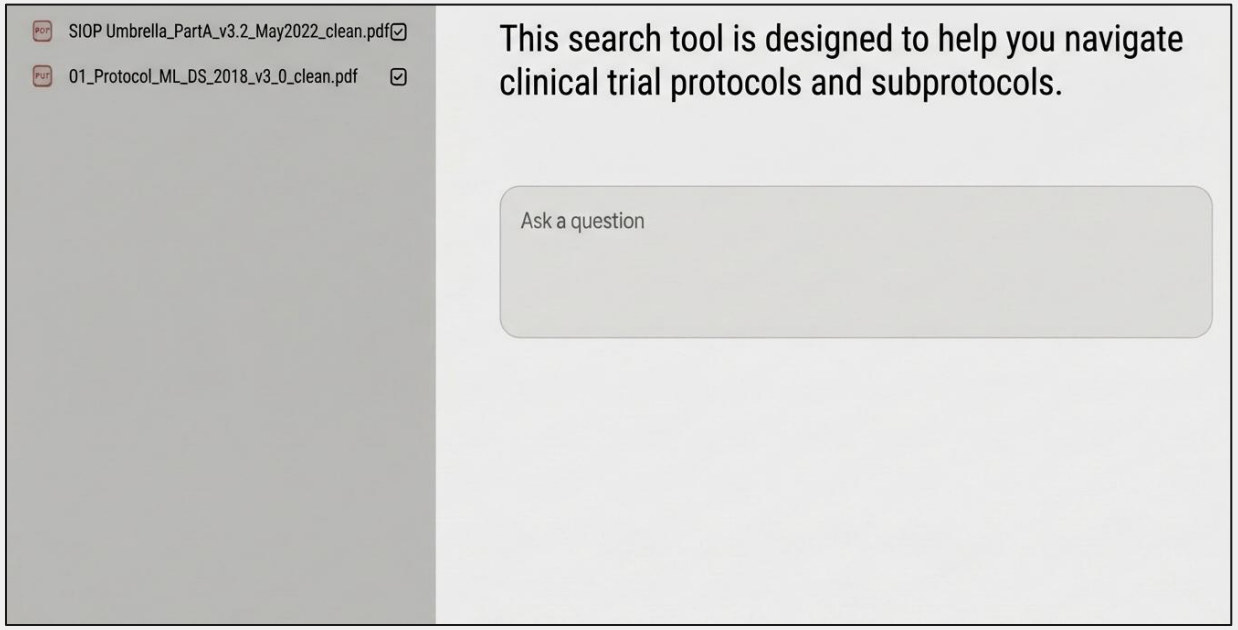
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> Tool to build: Must-haves Features & Requirements (2/2)

- **Context-Aware:** 2-tiered-context covering the applicable protocols and documentation

  a. Tier-1: **Upon Authentication from the BSPHO (Security Layer):** When a user from *UZ-Gent* logs in, the backend populates the left-hand menu with only the protocols applicable to *UZ-Gent*.
  Note: the auth and final integration to BSPHO portail are not covered in this hackathon, but the backend / UI should be built accordingly.
  Note: A mapping table (hospitals; protocols) has been provided.

  b. Tier-2: **Protocol Selection (UX Layer)**: The user can check / uncheck protocols in the left menu and asks questions on the selected protocols.



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> Tool to build: Hackathon scope

Given limited hackathon time, teams should focus on delivering the core value proposition: the primary search workflow, accurate grounding, and document ingestion capabilities:

- **Core MVP Feasibility:** Build a side-by-side chatbot panel with selectable sources and inline citations.
- **Document Management Workflow:** Develop a lightweight admin interface to upload new PDF protocols directly to GCS, select applicable hospitals / centers, then trigger automatic re-indexing.
- **Stretch Goals**
    a. **Should-have features:**
        i. **(Extensibility)** Beside the primary treatment protocols, there are supporting documents (eg: subprotocols), implement a "deep search" method to include these documents in the search results. Note: the UX left-side panel (with sources/protocols) can be leveraged.
        ii. **(Feedback)** Implement thumbs up/down UI components to provide feedback on the search tool.
    b. **Could-have features**
        i. **(Clicking links)** Clicking links to the cited source or PDF-viewer (Gemini Notebook UX style).
        ii. **(Multi-language support)** The tool could provide answers in Dutch.

‼ Out of Scope for Hackathon: Integration of the search tool into the BSPHO portal. Authentication to BSPHO portal.

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> Tool to build: Materials & Resources

- **Protocols & subprotocols: here**
- **Golden Dataset & Mapping table (Hospitals; Protocols): here**